A Complete Native Cut Archive at the First Unsupported Carrier

Exact Reconnection, Totality Defects, and an H1 Subgate Certificate

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Abstract

TPC-133–135 replace the abstract native-archive schema by an executable entrance, a complete dyadic prefix–tail compiler, and an exact eligible/frontier partition. We compose them into one growing cut archive. Every terminal path has exactly one type:

eligible prefix soft, eligible tail open, frontier unmapped.

No frontier occurrence is deleted or declared negligible. The composite preserves every native source, the prescribed shift, and one physical normalization, and satisfies

$$\mathbf{1}^T M_X^{\mathrm{cut}} = \mathbf{1}^T.$$

Hence the original hard packet is exactly the sum of the three typed terminal families; the published TPC-17 theorem makes only the first family soft. We then distinguish path-totality from operator-totality for the determinant-fiber, zero-mode-order, physical-grouping, and downstream fixed-shift maps. None is currently total on all nonsoft paths. Thus the native cut subgate is proved at L1, while the original actual-carrier H1 node remains NOT-TESTABLE. This is not a positive L2 result and proves no prime-pair theorem.

1 Three exact terminal types

TPC-133 produces every native tuple in a certified support envelope; TPC-134 gives every dyadic prefix–tail path; and TPC-135 divides the blocks into the published-core eligible set and its exact frontier [7–9]. We retain these source artifacts by content hash for reproducibility, while tuple and path identities themselves remain explicit mathematical records.

Definition 1.1 (Cut terminal type)**.** Every TPC-134 path receives exactly one label:

(i) EPS: its block is eligible and its divisor type is prefix;

(ii) ETO: its block is eligible and its divisor type is tail;

(iii) FUM: its block is in the frontier.

The labels mean eligible-prefix-soft, eligible-tail-open, and frontier-unmapped, respectively.

An EPS record includes a theorem source and the exact scope of the TPC-17 prefix estimate. The other two labels contain no soft theorem source. In particular, FUM is a retained mathematical leaf, not an error code.

Theorem 1.2 (Complete native cut archive)**.** *For every X, the composition above gives a finite path matrix M_X^{cut} with exactly one cut record for every TPC-134 path. It satisfies*

$$\boxed{\mathbf{1}^T M_X^{\mathrm{cut}} = \mathbf{1}^T}$$

and intertwines the native tuple, h_0, and the inherited physical normalization.

Proof. TPC-134 proves the column identity and metadata intertwining. TPC-135 partitions the block set, while the prefix/tail flag partitions every eligible block. Relabeling rows by the resulting three disjoint cases does not change any multiplier. It therefore preserves the column sums and all copied metadata. □

Corollary 1.3 (Exact cut reconnection)**.** *Writing $S_{\mathsf{EPS}}, S_{\mathsf{ETO}}, S_{\mathsf{FUM}}$ for the three terminal scalar sums,*

$$\boxed{B_{h_0,\delta}(X) = S_{\mathsf{EPS}}(X) + S_{\mathsf{ETO}}(X) + S_{\mathsf{FUM}}(X).}$$

On the fixed published-core scope, for every $A > 0$,

$$S_{\mathsf{EPS}}(X) = O_A(X(\log X)^{-A}),$$

and hence

$$B_{h_0,\delta}(X) = S_{\mathsf{ETO}}(X) + S_{\mathsf{FUM}}(X) + O_A(X(\log X)^{-A}).$$

Proof. The exact equality follows from theorem 1.2. Each EPS block lies in the scope of the cumulative divisor prefix theorem of TPC-17, and there are only $O_W(\log X)$ relevant blocks [10]. Nothing is estimated on the other types. □

2 Partial downstream maps

The cut archive is not yet the physical archive of TPC-124 or the fixed-shift archive of TPC-125 [4, 5]. Four maps must be distinguished:

$$Q_D, \qquad Q_Z, \qquad G, \qquad P_{h_0},$$

representing determinant fibers, canonical zero-mode order, physical grouping, and the downstream fixed-shift selector. The first two must match the literal data used by TPC-121 and TPC-122 [1, 2].

Definition 2.1 (Path-totality and operator-totality)**.** Let F be a partial map on terminal cut records and let M^{ret} consist of the nonsoft rows.

(a) F is *path-total* when every retained path with nonzero edge multiplier has terminal record in $\mathrm{dom}(F)$.

(b) With $P_{\mathrm{dom}(F)}$ the coordinate selector, F is *operator-total on the archive* when

$$(I - P_{\mathrm{dom}(F)})M^{\mathrm{ret}} = 0.$$

Here "nonzero" refers to the explicit dyadic edge multiplier, not to the full arithmetic coefficient of its native source. Possible zeros of W, μ, or r_Q are not used to delete an unmapped path.

Theorem 2.2 (Row-separated totality equivalence)**.** *For the row-separated path matrix M^{ret}, path-totality is equivalent to operator-totality as defined above. By contrast, the aggregated scalar condition*

$$\mathbf{1}^T(I - P_{\mathrm{dom}(F)})M^{\mathrm{ret}} = 0$$

is strictly weaker and is not a provenance certificate.

Proof. Under path-totality every retained row with nonzero edge multiplier is selected, so the displayed matrix defect vanishes. Conversely, if a retained path with nonzero edge multiplier is omitted, its row remains a nonzero row of $(I - P_{\mathrm{dom}(F)})M^{\mathrm{ret}}$; entries in a different row cannot cancel it. Thus a zero matrix defect implies path-totality.

For the last assertion, take a one-column omitted matrix with two rows

$$M^{\mathrm{ret}} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \qquad P_{\mathrm{dom}(F)} = 0.$$

Then $\mathbf{1}^T M^{\mathrm{ret}} = 0$, although $M^{\mathrm{ret}} \neq 0$ and both paths are omitted. This is the cancellation mechanism in aggregated local-defect checks of TPC-123 [3]; it does not occur inside the row-separated matrix equality. □

Corollary 2.3 (Current totality verdict). *In the committed TPC-136 map manifest, none of Q_D, Q_Z, G, P_{h_0} is path-total on the nonsoft cut archive. Consequently*

$$\boxed{\mathsf{H1.native\_cut} = \mathrm{PROVED}_{\mathrm{L1}},} \qquad \boxed{\mathsf{H1.actual\_carrier} = \textsc{not-testable}.}$$

Proof. The first statement is theorem 1.2. The map manifest lists no theorem-backed domain covering every ETO and FUM occurrence. Thus the actual-carrier fields required by the original MVP3 node cannot yet be evaluated [6]. □

For each of the four maps, the manifest separately records the path-totality contract, the inherited fixed value of h_0, and the status of its theorem source. All four map statuses and source statuses are currently NOT-TESTABLE; in particular, merely copying h_0 into cut metadata is not a theorem-backed construction of P_{h_0}.

Remark 2.4 (No movement of the goalpost). If a future route definition chooses the present cut as the terminal scope of H1, then the cut subgate is closed and the four total maps belong to downstream nodes. Under the existing MVP3 wording, "complete actual-carrier native archive," we do not silently make that change. The first missing item is now more precise: totalize the frontier, or prove its complete scalar soft in the inherited normalization, and then supply total Q_D, Q_Z, G, P_{h_0} maps.

3 Executable certificate and kill rules

The standard-library program `experiments/tpc136_cut_archive.py` binds the TPC-134 paths to the passing TPC-135 certificate, creates one cut record per path, and writes a downstream-map manifest. It reports missing nonsoft path counts separately for all four maps. It rejects deletion, duplication, terminal-type mutation, an unsupported soft label, a wrong shift, a false claim that an empty-domain map is total, a promoted source without a theorem, and a wrong fixed-h_0 source contract. Default mode writes deterministic artifacts; `-check` is strictly read-only.

Statement	Status
Complete archive through the declared native cut	Proved, L0/L1.
Eligible-prefix soft estimate	Imported in its exact published scope.
Frontier disposition	Retained and open; no scalar estimate.
Total determinant and zero-mode maps	Absent; NOT-TESTABLE.
Total physical and downstream shift maps	Absent; NOT-TESTABLE.
New positive fixed-h_0 arithmetic saving	None; no positive L2.

Kill criteria. A column defect, wrong native source, wrong h_0, second normalization, or unsupported soft tag stops the declared archive implementation. A map declared proved while omitting a retained path stops that map. A genuinely absent map remains NOT-TESTABLE, not architecture-infeasible.

Explicit exclusions. This paper proves no frontier cancellation, determinant-energy exponent, zero-mode exponent, complete physical cover, fixed-shift localization bound, endpoint below 1/400, parity breakthrough, Hardy–Littlewood asymptotic, prime-pair lower bound, or twin-prime theorem.

References

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